Instrumentation: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Basic telescope numbers. Return →

The focal ratio is $1000/200 = 5$, so f/5. Magnification is $1000/20 = 50\times$. Exit pupil is $200/50 = 4$ mm. **Check:** a 4 mm exit pupil is comfortable for dark-adapted visual observing.

Solution 2 — Diffraction and seeing. Return →

The Rayleigh limit is $138.4/200 = 0.692''$. Since the seeing is $1.5''$, the practical resolution is about $1.5''$, not $0.69''$. **Check:** the atmosphere usually dominates for modest ground-based apertures.

Solution 3 — Light grasp. Return →

Light grasp scales as area, so the ratio is $(200/50)^2 = 16$. The magnitude gain is $2.5 \log_{10} 16 = 3.01$ mag. **Check:** a fourfold aperture increase gives a large but not infinite gain; sky background and detector noise still matter.

Solution 4 — Pixel scale. Return →

The scale is $206.265 \times 5/1000 = 1.03''$/pixel. A 1.5″ seeing disk spans only about 1.5 pixels, so it is undersampled for accurate shape or precision photometry. **Check:** two to three pixels across the FWHM is a common practical target.

Solution 5 — Signal-to-noise. Return →

The total variance is $10000 + 2500 + 100 + 25 = 12625$. The noise is $\sqrt{12625} = 112.4$ electrons, so SNR $= 10000/112.4 = 89.0$. **Check:** the source dominates, but sky and read noise still reduce the ideal $\sqrt{10000} = 100$.

Solution 6 — Doubling the exposure. Return →

If source noise dominates, SNR $\propto \sqrt{t}$, so doubling time increases SNR by $\sqrt{2} = 1.414$. Doubling aperture quadruples collected source photons, so in the same source-dominated limit SNR doubles, not merely increases by $\sqrt{2}$. **Check:** aperture changes photon rate; exposure changes integration time.

Solution 7 — Trailing without tracking. Return →

The trail length is $15''/\text{s} \times 60\ \text{s} = 900'' = 15'$, a quarter of a degree. This is far larger than typical seeing or pixel scales, so stars become long streaks and photometry or astrometry is ruined. **Check:** even a few seconds of untracked exposure can trail noticeably at long focal length.

Solution 8 — Photon budget. Return →

The collecting area is $\pi(0.10)^2 = 0.0314\ \text{m}^2$. A 10th-magnitude star is 10^{-4} times the zero-magnitude flux. Detected electrons are $10^{10} \times 0.0314 \times 0.1 \times 10^{-4} \times 0.8 \times 10 = 2.51 \times 10^4$. **Check:** the answer is tens of thousands of electrons, plausible for a short small-telescope exposure.

Solution 9 — Calibration frames. Return →

Bias frames measure the electronic zero level. Dark frames measure thermal signal accumulated during an exposure. Flat frames measure relative pixel response, vignetting and dust shadows. The flat field is the correction that directly addresses pixel-to-pixel sensitivity and dust shadows. **Check:** calibration removes instrumental structure before scientific interpretation.

Solution 10 — Choose an instrument. Return →

A 150 mm telescope has a Rayleigh limit of $138.4/150 = 0.92''$, so it meets the diffraction requirement for a 1.0″ pair. But 1.0″ seeing is comparable to the separation, so a clean split is not guaranteed. Better seeing, adaptive optics, lucky imaging, interferometry or space observation would help more than magnification alone. Magnification enlarges the image but cannot restore information blurred by the atmosphere.